

Yili Wen

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Research Interests

I design and build **wearable and soft-robotic systems that restore and augment human movement**. I am drawn to the full loop from human intent to physical assistance, and I work across **soft actuation, wearable haptics, embedded sensing, and digital fabrication**. So far this has taken two forms: a soft pneumatic exoskeleton that helps people with spinal cord injury grasp, and a haptic wearable that guides blind and low-vision runners — two applications of one question, how a garment on the body can stand in for a function the body has lost. My approach is hands-on and human-centered: I prototype real hardware and evaluate it with the people it is built for. Looking ahead, I want to extend this work toward **assistive and rehabilitation exoskeletons** and toward closing the loop between neural intent and adaptive wearable assistance.

Education

- 2025 – Present **Graduate Student, Computer Science — The University of Texas at Dallas**
Design & Engineering for Making (DE4M) Lab, advised by Liang He.
M.S. in Computer Science (Interactive Computing) expected **Spring 2027**.
- 2021 – 2025 **B.A. in Computation and Design — Duke Kunshan University**
Dual-degree program with **B.A. in Interdisciplinary Studies, Duke University**.
Undergraduate research advisor: Xin Tong.

Publications

* *First author.* *Full author lists preserved; names abbreviated where long.*

- 2026 [E.1] ***Yili Wen**, Yafei Ge, Xin Tong, Liang He. “PufFab: Making Shape-Transformable Rice Paper for Playful Food Fabrication.” (Work-in-Progress). *ACM TEI 2026*.
- 2026 [W.2] ***Yili Wen**, Ceci Verbaarschot, Liang He. “From Intention to Action: An Invasive BCI-Driven Wearable Framework for Everyday Hand Assistance in SCI Patients.” (Workshop paper & poster). *Everyday Wearable for Personalized Health and Well-Being Workshop, ACM CHI 2026*.
- 2026 [P.1] Georgia Loewen, Karen Anne Cochrane, **Yili Wen**, Audrey Girouard. “Personas at Play: Understanding Disabled Player Experience to Expand Accessible Controller Design.” (Poster). *Graphics Interface (GI) 2026*.
- 2025 [C.1] ***Yili Wen**, Jiaxin Wang, Hongni Ye, Liang He, Min Fan, Xin Tong. “‘I Want to Use Technology to Help Farmers in the Future’: Enhancing Children’s Understanding of Farming through Tangible User Interfaces and Embodied Interaction.” *ACM Interaction Design and Children (IDC) 2025*.
- 2024 [C.2] Hongni Ye, Ruoxin You, Kaiyuan Lou, **Yili Wen**, Xin Yi, Xin Tong. “‘I Keep Sweet Cats in Real Life, But What I Need in the Virtual World Is a Neurotic Dragon’: Virtual Pet Design with Personality Patterns.” *Chinese CHI 2024*.
- 2023 [W.1] Hongni Ye, Ruoxin You, Kaiyuan Lou, **Yili Wen**, Xin Yi, Xin Tong. “PetGen: Design and Generation of Virtual Pets.” (Workshop). *IEEE ICME Workshops (ICMEW) 2023*.

Manuscripts In Preparation

- Tianjian Liu, Jingwei Sun, **Yili Wen**, Haiqi Shen, Liuxin Zhang, Yu Zhang, Qianying Wang, Xin Tong. “Human-AI Co-Writing in Knowledge-Intensive Tasks: Behaviors, Challenges, and Collaboration Patterns.” In preparation.

Selected Research Projects

2025 – Present	Soft Grasp-Assist Exoskeleton for Spinal Cord Injury (“From Intention to Action”) A wearable soft exoskeleton that restores grasping to people with spinal cord injury, driven by the wearer’s own decoded intent. - Designed soft pneumatic finger actuators (silicone-cast) and a layer-jamming wrist brace for stable, compliant grasp assistance during everyday tasks. - Built the embedded control and pressure-sensing stack, and a neural-intent (BCI) pipeline that turns decoded intention into adaptive assistance. - Ground the design in real rehabilitation needs through a co-design study with clinicians.
2025 – Present	Haptic Wearable for Blind & Low-Vision Runners A body-worn haptic system that guides blind and low-vision runners through turns and hazards without relying on audio. - Built a belt of distributed skin-stretch actuators that deliver directional cues on the body, paired with on-device vision and a companion iOS app. - Investigating a novel guidance mechanism — sequential (cranio-caudal) vs. simultaneous cueing — evaluated in the field with blind and low-vision runners.
2025	PuffFab: Shape-Transformable Rice Paper for Playful Food Fabrication (TEI ’26) A design-to-fabrication pipeline for edible materials that morph from flat prints into 3D forms. Modified 3D-printer hardware and built the toolchain to print 2D patterns that transform into 3D structures when fried.
2024 – 2025	Adaptive Game-Controller Hardware for Upper-Limb Motor Disabilities (GI ’26) DIY, reconfigurable game controllers that let players with upper-limb motor disabilities shape the input to their own body. - Designed and fabricated the modular, remappable controller prototypes that served as the physical platform for the user study. - Co-authored the resulting poster (Graphics Interface 2026), which studied player experience and narrative-inquiry methods around this platform.
2023 – 2024	WooGu: Tangible System for Children’s Agricultural Literacy (IDC ’25) An educational tangible system that teaches children about farming through hands-on, multi-sensory interaction. Designed and built the system, pairing 3D-printed farming tools with embedded sensors to give visual, haptic, and digital feedback.

Research Experience

2025 – Present	Research Assistant — Liang He Design & Engineering for Making (DE4M) Lab, UT Dallas, USA.
2023 – 2025	Research Assistant — Xin Tong HCI Lab, Duke Kunshan University & Information Hub, HKUST (Guangzhou), China.
2023 – 2025	Research Assistant — Karen Cochrane Embodied Computing Lab, University of Waterloo, Canada.
2022 – 2023	Research Assistant — Fan Liang Center for the Study of Contemporary China, Duke Kunshan University, China.

Teaching & Outreach

Summer 2026	Instructor — TAST STEM Bridge Summer Camp Department of Computer Science, UT Dallas. Six-week camp (June–July 2026); designed and delivered hands-on lectures and build sessions.
Summer 2026	Instructor — CS Outreach Summer Camp Department of Computer Science, UT Dallas. June–July 2026.
Spring 2026	Teaching Assistant — CS 4376 Object-Oriented Design Department of Computer Science, UT Dallas.

Technical Skills

- **Fabrication & Prototyping:** soft-robotics fabrication (silicone molding/casting), SLA (resin) & FDM 3D printing, OpenSCAD / CAD, laser cutting.
- **Electronics & Embedded:** ESP32, Arduino, soft pneumatic actuation & control, brushed DC motor control (H-bridge drivers), closed-loop encoder feedback, I²C sensor integration, ESP-NOW wireless mesh.
- **Machine Learning & Vision:** PyTorch, YOLO / Ultralytics (custom object-detection training), Core ML on-device deployment, OpenCV.
- **Programming:** Python, C/C++ (embedded), Swift/SwiftUI, JavaScript/p5.js, Java/Processing, R.
- **Design & Analysis:** Figma, Blender, statistical analysis in R.
- **Research Methods:** controlled & field user studies, mixed-methods, qualitative coding, sensor-data analysis.

Presentations

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| Apr 2026 | Poster & Talk — “From Intention to Action: An Invasive BCI-Driven Wearable Framework for Everyday Hand Assistance in SCI Patients”
Everyday Wearable for Personalized Health and Well-Being Workshop, ACM CHI 2026, Barcelona, Spain. |
| 2026 | Talk & Poster — “PuffFab: Making Shape-Transformable Rice Paper for Playful Food Fabrication”
ACM TEI 2026. |
| 2025 | Paper Presentation — “‘I Want to Use Technology to Help Farmers in the Future’ ”
ACM Interaction Design and Children (IDC) 2025. |

Awards

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| 2023 | Center for the Study of Contemporary China (CSCC) Fellowship
Duke Kunshan University. |
| 2022 – 2024 | Dean’s List
Duke Kunshan University (Fall 2022, Fall 2023, Spring 2024, Fall 2024). |